

Integrated Development Project

2025 / 2026

Specialization: Data Science

**Project Title: AI-Powered Intelligent Interview &
Recruitment Platform**



Prepared by:

CHAYMA BEN CHEKAYA

AKREMI RAOUNEK

NESRINE FELAH

SAMAA ZILI

GHARBI MALEK

MOUHAMED AMINE MANSOUR

Academic Supervisor:

Nafaa Haffar – Abdelkarim Mars

Contents

General Introduction	6
1 General Context	8
1.1 Introduction	8
1.2 Project Overview	8
1.2.1 Project Objectives	9
1.3 Task Distribution Within the Team	9
1.4 Conclusion.....	10
2 System Design and AI Model Architecture	11
2.1 Introduction	11
2.2 CV Classification & Matching Module	11
2.2.1 Dataset	11
2.2.2 Model Approach	12
2.2.3 Choice of Models for CV Classification and Matching.	13
2.2.4 Matching CV and Job Offers	14
2.2.5 Proposed Approach	14
2.3 CV Analysis & Guidance Module	14
2.3.1 Dataset	14
2.3.2 Model Approach	16
2.3.3 Choice of Model	17
2.3.4 Proposed Approach	17
2.4 Career Recommendation Module	17
2.4.1 Dataset	18
2.4.2 Model Approach	19
2.4.3 Choice of Recommendation Model	19
2.4.4 Proposed Approach	19
2.5 Gesture Analysis Module	19
2.5.1 Dataset	19
2.5.2 Model Approach	20
2.5.3 Choice of Model	21



2.5.4	Proposed Approach.....	21
2.6	Audio Analysis Module	21
2.6.1	Dataset.....	21
2.6.2	Model Approach.....	22
2.6.3	Choice of Model.....	23
2.6.4	Proposed Approach.....	23
2.7	Sentiment Analysis Module	23
2.7.1	Dataset.....	23
2.7.2	Model Approach.....	24
2.7.3	Choice of Model.....	25
2.7.4	Proposed Approach.....	25
2.8	Conclusion.....	25
3	Training and Model Evaluation	26
3.1	Introduction	26
3.2	Data Preparation and Processing	26
3.3	Tools and Libraries Used	27
3.4	Software Architecture	27
3.5	Evaluation of the Models.....	28
3.5.1	Model Evaluation Overview	28
3.5.2	Overall Model Performance.....	28
3.5.3	Discussion of Results	29
3.6	Conclusion.....	29
4	Deployment and Real-Time Testing	30
4.1	Introduction	30
4.2	Real-Time System Testing	30
4.2.1	User Dashboard Interfaces	30
4.2.2	CV Classification & Matching Test.....	32
4.2.3	CV Analysis & Guidance Module Test.....	32
4.2.4	Career Recommendation Module Test	33
4.3	Conclusion.....	34
	General Conclusion and Perspectives	35
	Perspectives	35

List of Figures

2.1	General Architecture of the CV Classification and Matching System . .	13
2.2	General Architecture of the CV Analysis & Guidance Module.....	16
2.3	General Architecture of the Career Recommendation Module.....	18
2.4	General Architecture of the Gesture Analysis Module.....	20
2.5	General Architecture of the Audio Analysis Module.....	22
2.6	General Architecture of the Sentiment Analysis Module.....	24
4.1	Candidate Dashboard Interface.....	31
4.2	Recruiter Dashboard Interface.....	31
4.3	CV Classification and Job Matching Interface — Real-Time Test.....	32
4.4	CV Analysis & Improvement Suggestions Output.....	33
4.5	Career Recommendation Results Interface.....	34

List of Tables

- 1.1 Task Distribution Among Team Members 10
- 2.1 Dataset Distribution — CV Classification Module..... 12
- 2.2 Comparison of CV Classification Models..... 13
- 2.3 CV–Job Matching Methods Comparison 14
- 2.4 Dataset Distribution — CV Analysis Module 15
- 2.5 Comparison of CV Analysis Models..... 17
- 2.6 Dataset Description — Career Recommendation Module 18
- 2.7 Comparison of Recommendation Approaches..... 19
- 2.8 Dataset Description — Gesture Analysis Module 20
- 2.9 Comparison of Gesture Analysis Approaches..... 21
- 2.10 Dataset Description — Audio Analysis Module..... 22
- 2.11 Comparison of Audio Analysis Models..... 23
- 2.12 Dataset Description — Sentiment Analysis Module..... 24
- 2.13 Comparison of Emotion Recognition Models 25
- 3.1 Tools and Libraries Used in TalentSphere 27
- 3.2 Global Performance of TalentSphere AI Models..... 28

General Introduction

In today's increasingly competitive job market, traditional recruitment processes present significant limitations for both candidates and recruiters. Candidates, particularly students and recent graduates, often struggle to find opportunities suited to their profiles. They send numerous applications without receiving responses, and their CVs are not always adequately analyzed despite their qualifications. On the other hand, recruiters face a high volume of applications, making the selection process time-consuming and complex. It becomes difficult to quickly identify the most relevant profiles, leading to time loss and inefficiency in the recruitment process.

To address these challenges, we have developed **TalentSphere**, an intelligent platform aimed at simplifying, automating, and improving the recruitment process. The platform offers two main spaces: a *Candidate Space* and a *Recruiter Space*.

In the **Candidate Space**, users can create an account, browse and apply for job offers, and submit their CV. The platform also features a quiz system adapted to each job offer, allowing candidates to assess their skills and generate a score. Additionally, candidates receive personalized advice to improve their CV based on their targeted job offer. **TalentSphere** also provides a job recommendation system based on the candidate's profile to suggest the most relevant opportunities. An intelligent chatbot is integrated to simulate interviews, allowing candidates to practice and evaluate their communication, technical, and behavioral skills, with feedback in the form of scores and analysis of strengths and weaknesses.

In the **Recruiter Space**, users can publish job offers and access a list of applications automatically ranked according to several criteria, including quiz scores and CV-offer compatibility. This allows for rapid selection of the most qualified profiles. The platform also features an online interview system, including analysis of candidate behavior such as facial expressions, voice, and communication. At the end of the interview, a detailed report is generated and sent to the recruiter, compiling the candidate's scores (quiz, interview, and matching with the offer).

In summary, **TalentSphere** is a comprehensive solution that digitizes and optimizes recruitment by combining automation, intelligent assessment, and personalized support



to create a better experience for both candidates and recruiters.

This report is structured as follows. **Chapter 1** presents the general context of the project, including the objectives, team structure, and task distribution. **Chapter 2** describes the design and architecture of all AI models used across the different modules of the platform. **Chapter 3** covers the training process, evaluation methodology, and performance results for each model. Finally, **Chapter 4** details the deployment of the platform and presents real-time testing results to validate the system under realistic conditions.

General Context

1.1 Introduction

This first chapter presents the general context of our project, **TalentSphere**, an intelligent platform designed to optimize and digitize the recruitment process. In today's highly competitive job market, candidates often struggle to access opportunities that match their profiles, while recruiters face difficulties in efficiently handling a large volume of applications.

In this context, **TalentSphere** aims to improve the overall recruitment workflow by integrating artificial intelligence technologies to automate candidate screening, evaluate skills, and provide a smoother and more personalized experience for both candidates and recruiters. This chapter also introduces the main objectives of the project and its overall structure.

1.2 Project Overview

TalentSphere is an intelligent recruitment platform that connects candidates and recruiters using automation and AI. The platform has two main parts:

- **Candidate Space:** Users can create accounts, apply for jobs, upload CVs, take skill quizzes, and receive career recommendations. An AI chatbot also simulates interviews and provides feedback.
- **Recruiter Space:** Recruiters can manage applications more easily, with candidates automatically ranked based on CV matching, quiz results, and behavior analysis. The platform also offers online interviews with detailed evaluation reports.



1.2.1 Project Objectives

The main objective of **TalentSphere** is to modernize and optimize the recruitment process by making it smarter, faster, and more objective. More specifically, the project aims to:

1. Facilitate access to job opportunities for students and recent graduates.
2. Help recruiters quickly identify the most relevant candidate profiles.
3. Reduce the time required to process applications through automation.
4. Improve the quality of candidate evaluation using AI-based tools.
5. Provide a personalized experience through intelligent recommendations and detailed feedback.

Note

In summary, **TalentSphere** offers an innovative solution that transforms traditional recruitment into an intelligent, data-driven, and skill-oriented process.

1.3 Task Distribution Within the Team

To ensure an effective organization of the project, tasks were clearly distributed among team members based on the different modules of the **TalentSphere** platform.

**Table 1.1.** Task Distribution Among Team Members

Member	Main Task	Task Description
Chayma	CV Classification & Matching	Develop the system for classifying CVs and matching candidates with job offers.
Sama	Career Recommendation	Design an AI-based system that suggests suitable career paths and opportunities.
Nesrine	CV Analysis & Guidance	Analyze CV content and provide personalized feedback and improvement suggestions.
Rawnek	Gesture Analysis	Implement gesture recognition to evaluate candidate behavior during interviews.
Amine	Sentiment Analysis	Develop a system to analyze candidate emotions and responses during interviews.
Malek	Audio Analysis	Process and analyze voice data to assess communication skills and tone.

1.4 Conclusion

This first chapter established the general framework of the **TalentSphere** project by presenting its objectives and overall approach to improving the recruitment process using artificial intelligence. It also highlighted the platform's main features and its two core spaces: candidate and recruiter.

The clear distribution of tasks within the team ensures effective organization and smooth progress across all project modules, from CV analysis and matching to behavioral evaluation and recommendation systems. This structure lays the foundation for the following chapters, which will detail the system design, implementation, and evaluation of the platform.

System Design and AI Model Architecture

2.1 Introduction

This chapter focuses on the design and architecture of the artificial intelligence models used in the **TalentSphere** platform. Unlike traditional systems based on a single model, **TalentSphere** relies on multiple specialized models, each dedicated to a specific module in the recruitment process.

First, we present the data used for each module, including CV data, text inputs, video streams, and audio recordings. Then, we justify the choice of different AI techniques such as machine learning, natural language processing (NLP), and deep learning, based on their effectiveness in similar applications. Finally, this chapter describes the architecture of the proposed system by detailing each model, its role, and how these models interact to ensure accurate, efficient, and intelligent recruitment processes.

2.2 CV Classification & Matching Module

2.2.1 Dataset

For this module, we used a recent dataset from Kaggle called *“Updated Resume Dataset”*, which is widely used for research in CV classification and recruitment automation. The dataset contains resumes classified into 25 different job categories, allowing the model to learn how to identify and match candidate profiles with relevant job offers.

To ensure effective training and evaluation, the dataset was divided into two main sets:



1. **Training set:** 2,385 resumes
2. **Test set:** 597 resumes

The dataset was transformed using the TF-IDF technique, resulting in 35,378 features, which helps convert textual CV data into numerical representations suitable for machine learning models.

Table 2.1. Dataset Distribution — CV Classification Module

Set	Samples	Description
Training	2,385	Used for model learning
Test	597	Used for evaluation
Total	2,982	Full dataset

2.2.2 Model Approach

To improve classification accuracy and robustness, we adopted a hybrid approach combining multiple models:

1. **XGBoost Model:** The primary model used for CV classification based on TF-IDF features. It provides fast and efficient predictions.
2. **LLM (Large Language Model):** Acts as a complementary system. When XGBoost produces a low-confidence prediction, the LLM takes over to refine the classification using deeper contextual understanding.
3. **SpaCy:** Used for NLP tasks such as entity recognition and keyword extraction, helping improve classification and matching accuracy.

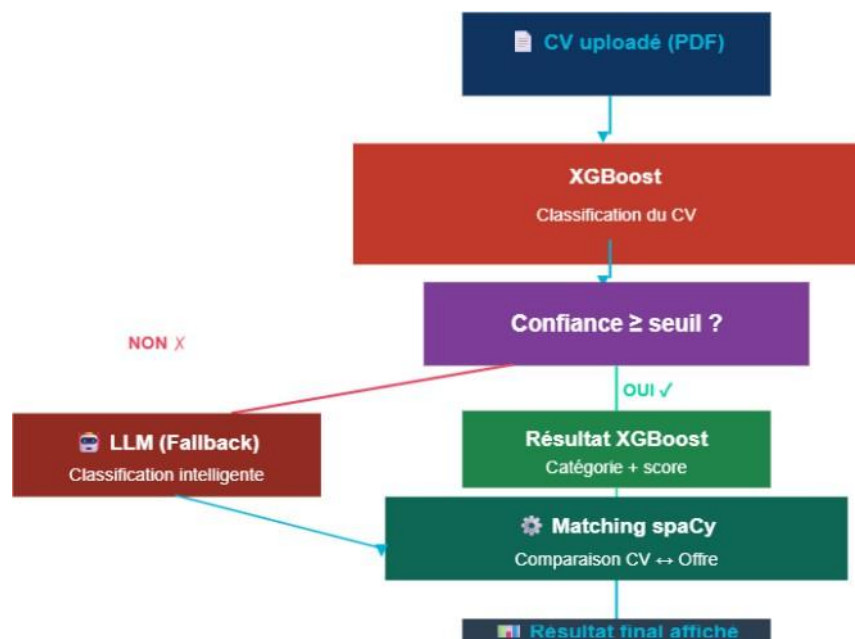


Figure 2.1. General Architecture of the CV Classification and Matching System

2.2.3 Choice of Models for CV Classification and Matching

The main objective of this module is to automatically identify the most relevant job category for each candidate and ensure accurate matching between CVs and job offers. To address this task, different approaches from Machine Learning (ML) and Deep Learning (DL) have been studied. A comparative analysis is presented in Table 2.2.

Table 2.2. Comparison of CV Classification Models

Method	Dataset Size	Accuracy	Latency	Complexity
Random Forest	~1K CVs	~70%	Fast	Low
XGBoost	962 CVs	~96%	Fast	Low–Med
BERT / CamemBERT	13K CVs	~92%	Slow	Very High
LLM (Gemma / LLaMA)	13K CVs	~92–95%	Very Slow	Very High



2.2.4 Matching CV and Job Offers

For the matching task between CVs and job descriptions, several NLP approaches have been analyzed. A comparison of matching techniques is presented below.

Table 2.3. CV–Job Matching Methods Comparison

Method	Dataset	Performance	Latency
spaCy NER	Resume datasets	~85–90%	Very fast
Custom spaCy NER	ResumeNER datasets	~80–90%	Fast
BERT-based NER	22K CVs	~90% F1	Medium

2.2.5 Proposed Approach

Based on the comparative study, we propose a hybrid intelligent architecture:

1. **XGBoost + TF-IDF** → primary fast classifier for CV categorization
2. **LLM (fallback system)** → used when XGBoost confidence is low, improving semantic understanding
3. **SpaCy NLP pipeline** → for entity extraction (skills, experience, education) and structured information parsing

2.3 CV Analysis & Guidance Module

2.3.1 Dataset

For this module, we used the Kaggle dataset “*UpdatedResumeDataSet*”, widely used for resume analysis and recruitment systems. The dataset was divided as follows:



Table 2.4. Dataset Distribution — CV Analysis Module

Set	Proportion	Description
Training	80%	Used for model fine-tuning
Test	20%	Used for evaluation

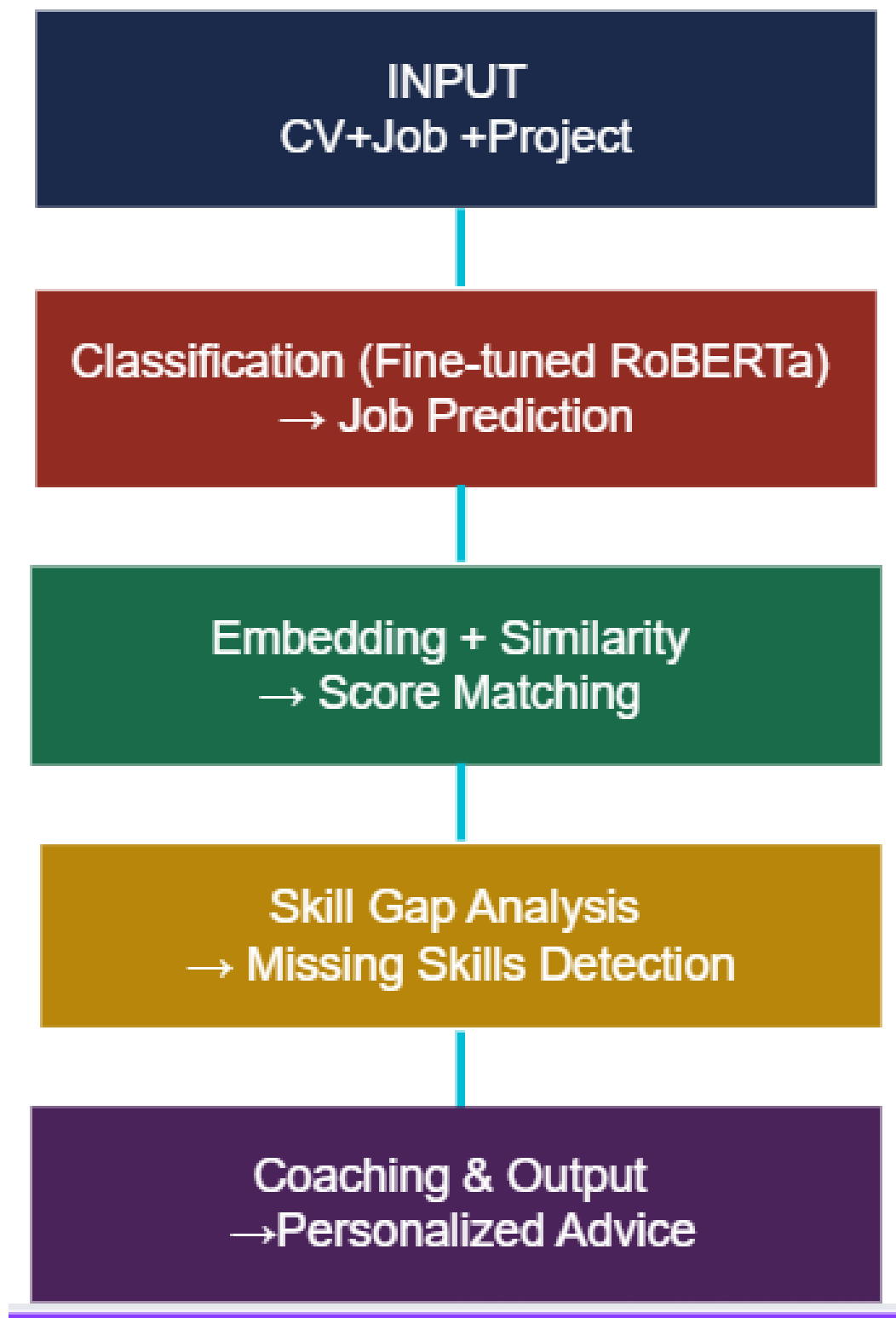


Figure 2.2. General Architecture of the CV Analysis & Guidance Module

2.3.2 Model Approach

To analyze CVs and generate personalized feedback, we adopted a fine-tuned transformer-based model: **RoBERTa**, pre-trained and fine-tuned on the resume dataset to under-



stand semantic structure, skills, experience, and job relevance.

The model is designed to:

1. Analyze CV structure and content
2. Detect missing or weak sections
3. Identify relevant skills and experiences
4. Generate personalized improvement suggestions

2.3.3 Choice of Model

Table 2.5. Comparison of CV Analysis Models

Rule-based systems	Small datasets	Limited	Very fast
TF-IDF + ML	1K–5K CVs	70–85%	Fast
BERT / CamemBERT	Large datasets	~90%	Slow
RoBERTa (Proposed)	UpdatedResumeDataSet	High	Medium

2.3.4 Proposed Approach

1. **RoBERTa fine-tuned** → main engine for deep CV understanding
2. **NLP preprocessing pipeline** → cleaning, tokenization, structure extraction
3. **Feedback generation module** → converts model predictions into human-readable advice

2.4 Career Recommendation Module



2.4.1 Dataset

For this module, we used real-world job data from LinkedIn, combined with a structured job_skills dataset that maps professional skills to job categories.

Table 2.6. Dataset Description — Career Recommendation Module

Dataset	Content	Purpose
LinkedIn Jobs	Job descriptions & titles	Training recommendation model
job_skills	Skill-to-job mappings	Skill-based matching enhancement

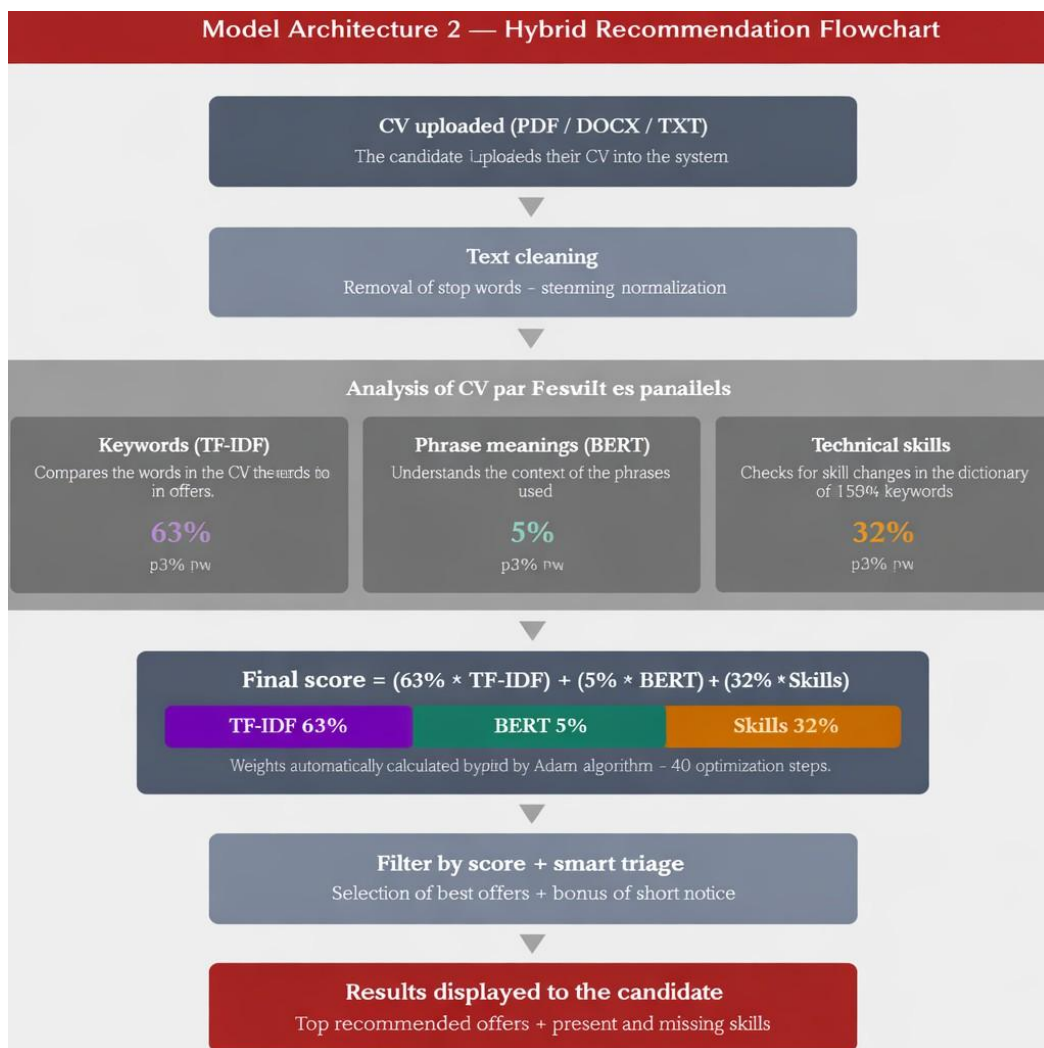


Figure 2.3. General Architecture of the Career Recommendation Module



2.4.2 Model Approach

We adopted a hybrid approach combining:

1. **TF-IDF:** Converts job descriptions and CV text into numerical representations.
2. **BERT:** Captures deep semantic relationships between skills and job descriptions.
3. **Skills-based Matching:** Extracts and compares candidate skills with required job skills.

2.4.3 Choice of Recommendation Model

Table 2.7. Comparison of Recommendation Approaches

Method	Dataset	Accuracy	Latency
TF-IDF only	Job descriptions	Medium	Very fast
Skill-based	job_skills	Rule-based	Very fast
BERT-based	Job datasets	High	Slow
Hybrid (Proposed)	LinkedIn + job_skills	High	Medium

2.4.4 Proposed Approach

1. **TF-IDF Module** → captures keyword importance in job descriptions
2. **BERT Model** → understands semantic similarity between CVs and jobs
3. **Skills Matching Engine** → ensures alignment between candidate skills and requirements

2.5 Gesture Analysis Module

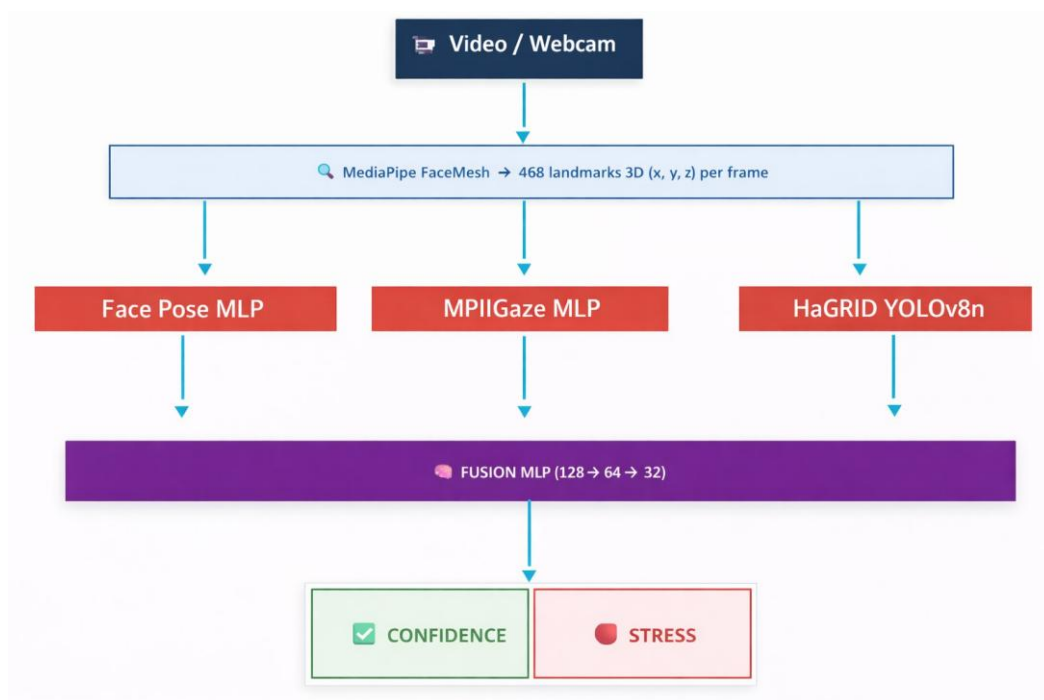
2.5.1 Dataset

Three complementary datasets were used to capture different aspects of candidate behavior:

**Table 2.8.** Dataset Description — Gesture Analysis Module

Dataset	Content	Purpose
Face Pose	Facial orientation images	Head pose estimation
MPIIGaze	Eye gaze tracking data	Eye movement analysis
HaGRID	Hand gesture images/videos	Gesture recognition

2.5.2 Model Approach

**Figure 2.4.** General Architecture of the Gesture Analysis Module

The system is based on a parallel multi-model architecture:

1. **FacePoseMLP:** Estimates head orientation and facial direction.
2. **MPIIGazeMLP:** Analyzes eye gaze direction and attention level.
3. **HaGRIDYOLOv8n + MLPFusion:** YOLOv8n for real-time detection, followed by an MLP classifier for gesture interpretation.

All three models operate in parallel, and their outputs are combined using a fusion MLP layer to produce a final behavioral evaluation score.



2.5.3 Choice of Model

Table 2.9. Comparison of Gesture Analysis Approaches

Method	Dataset	Performance	Latency
CNN-based models	Single dataset	Medium	Medium
MPIIGaze models	MPIIGaze	High (gaze)	Fast
YOLOv8n	HaGRID	High (detection)	Real-time
Fusion MLP (Proposed)	3 datasets	Very High	Medium

2.5.4 Proposed Approach

1. **Face Pose MLP** → head orientation analysis
2. **MPIIGaze MLP** → eye tracking and attention detection
3. **YOLOv8n + MLP (HaGRID)** → gesture recognition
4. **Fusion MLP layer** → combines all outputs into a behavioral score

2.6 Audio Analysis Module

2.6.1 Dataset

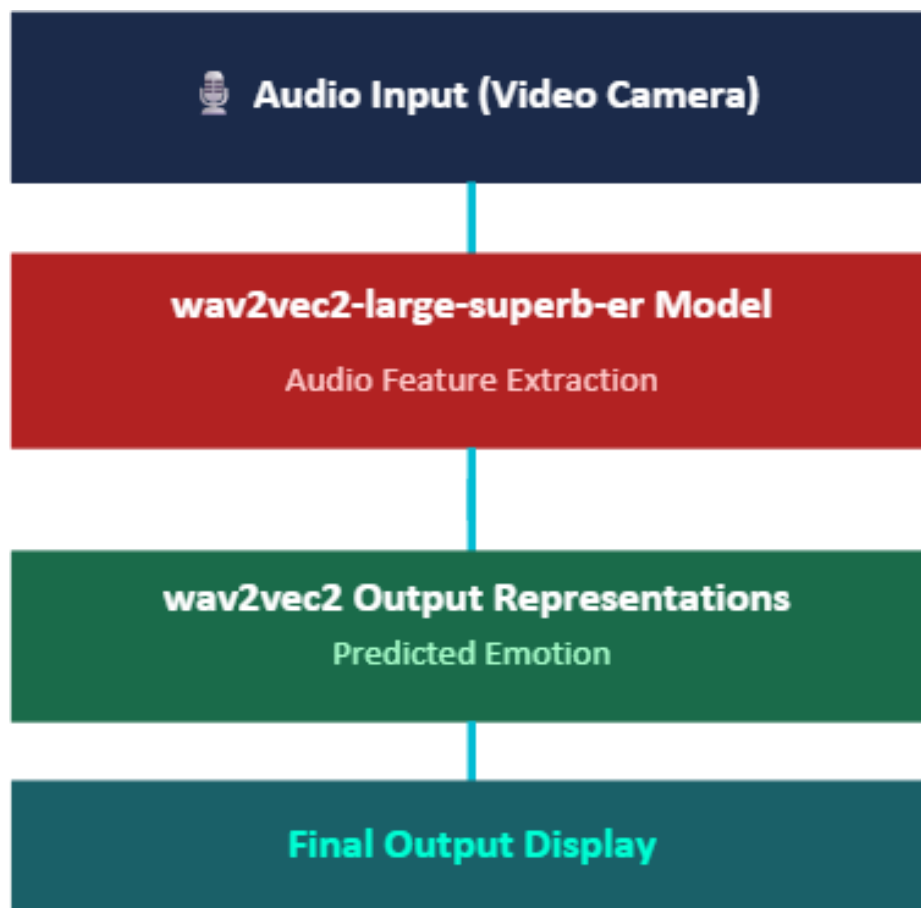
Two well-known speech emotion datasets were used:

1. **Toronto Emotional Speech Set (TESS)**
2. **CREMA-D dataset**

**Table 2.10.** Dataset Description — Audio Analysis Module

Dataset	Content	Purpose
TESS	Emotional speech recordings	Emotion classification
CREMA-D	Actor-based emotional speech	Speech emotion recognition

2.6.2 Model Approach

**Figure 2.5.** General Architecture of the Audio Analysis Module

The model used is **wav2vec2-large-superb-er**, a transformer-based model designed for speech emotion recognition that processes raw audio directly. It is used to:

- Detect emotional state of the candidate
- Analyze tone, confidence, and stress level



- Evaluate communication behavior during interviews

2.6.3 Choice of Model

Table 2.11. Comparison of Audio Analysis Models

Method	Dataset	Performance	Latency
MFCC + SVM	Small datasets	Medium	Very fast
CNN Audio Models	Speech datasets	High	Medium
LSTM-based models	Sequential audio	Med-High	Slow
wav2vec2 (Proposed)	TESS + CREMA-D	Very High	Medium

2.6.4 Proposed Approach

1. **wav2vec2-large-superb-er** → main model for emotion detection
2. **TESS + CREMA-D** → training and evaluation data
3. **Audio preprocessing pipeline** → normalization and noise reduction

2.7 Sentiment Analysis Module

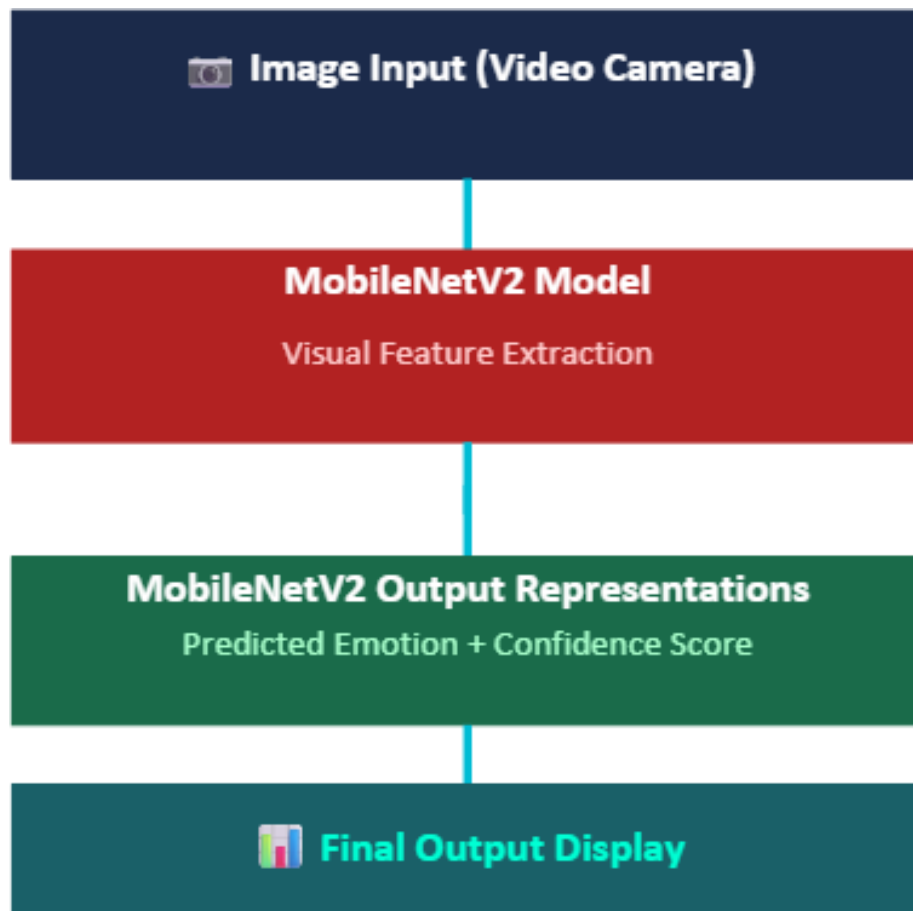
2.7.1 Dataset

For this module, we used the **FER2013 dataset** (Facial Expression Recognition 2013), containing grayscale facial images labeled with 7 emotional states: Happy, Sad, Angry, Neutral, Fear, Surprise, and Disgust.

**Table 2.12.** Dataset Description — Sentiment Analysis Module

Dataset	Content	Purpose
FER2013	Facial expression images	Emotion classification

2.7.2 Model Approach

**Figure 2.6.** General Architecture of the Sentiment Analysis Module

The model used is **MobileNetV2**, a lightweight convolutional neural network (CNN) designed for efficient image classification. It is used to:

1. Detect facial emotions from images
2. Analyze candidate emotional reactions during interviews
3. Support behavioral evaluation in the recruitment process



2.7.3 Choice of Model

Table 2.13. Comparison of Emotion Recognition Models

Method	Dataset	Performance	Latency
CNN basic models	FER2013	Medium	Medium
VGG16 / ResNet	FER2013	High	Slow
Deep CNN models	Large datasets	Very High	Slow
MobileNetV2 (Proposed)	FER2013	High + Efficient	Fast

2.7.4 Proposed Approach

1. **MobileNetV2** → main model for emotion classification
2. **FER2013 dataset** → training and evaluation data
3. **Image preprocessing pipeline** → normalization, resizing, augmentation

2.8 Conclusion

This chapter presented the design and architecture of the **TalentSphere** platform, which is based on multiple AI models working together across different modules. Instead of using a single model, the system combines specialized models for CV analysis, job recommendation, CV feedback, gesture analysis, speech analysis, and emotion detection.

Each module was built using the most suitable techniques, including transformer models (RoBERTa, BERT, wav2vec2), classical machine learning (XGBoost, TF-IDF), and deep learning models (MobileNetV2, YOLOv8n, MLP). This ensures good performance, accuracy, and real-time processing.

Training and Model Evaluation

3.1 Introduction

This chapter presents the training and evaluation process of the different AI models used in the **TalentSphere** platform. It first describes the data preparation process for the CV classification and matching module, including cleaning, preprocessing, and data augmentation techniques. Then, it introduces the tools and technologies used. Finally, it explains the implementation process and the methods used to train, test, and evaluate the performance of the AI models.

3.2 Data Preparation and Processing

For the CV Classification & Matching module, we used a structured dataset containing resumes categorized into multiple job domains. Before training, the data underwent several preprocessing steps:

1. Removal of missing and inconsistent data
2. Text normalization (CVs and category labels)
3. Text cleaning (removing noise and irrelevant characters)
4. Conversion of CV text into structured NLP-ready data

To improve model performance and reduce class imbalance, a text data augmentation technique was applied. This method generates synthetic CV samples by combining sentences from documents belonging to the same category. This augmentation helps to:



1. Balance class distribution
2. Enrich vocabulary diversity
3. Improve model robustness to real-world variations

3.3 Tools and Libraries Used

The development of the **TalentSphere** system relies on a combination of AI, backend, and frontend technologies.

Table 3.1. Tools and Libraries Used in TalentSphere

Tool / Library	Description	Usage in Project
Google Colab	Cloud-based Python environment	Model training and experimentation
React.js	Frontend JavaScript library	Candidate and recruiter user interface
NestJS	Node.js backend framework	API development and server-side logic
PostgreSQL	Relational database system	Storage of users, CVs, and results
Flask	Lightweight Python framework	AI model deployment via REST APIs
VS Code	Code editor	Full project development environment

3.4 Software Architecture

The **TalentSphere** system is built using a full-stack architecture combining web development and artificial intelligence:



- **Backend and APIs:** NestJS handles business logic, user management, and system communication. Flask deploys AI models as REST APIs.
- **Frontend:** React.js provides an interactive interface for both candidates and recruiters.
- **Database:** PostgreSQL stores CVs, job offers, user profiles, and model outputs.

3.5 Evaluation of the Models

3.5.1 Model Evaluation Overview

In this section, we present the evaluation results of all AI models used in the **TalentSphere** platform. Each module was tested separately using its corresponding dataset and performance metrics, mainly accuracy.

3.5.2 Overall Model Performance

Table 3.2. Global Performance of TalentSphere AI Models

Module	Model / Method	Accuracy
CV Classification & Matching	XGBoost + LLM + SpaCy	92%
Career Recommendation	TF-IDF + BERT + Skills Hybrid	91.29%
CV Analysis & Guidance	RoBERTa (fine-tuned)	93.48%
Gesture — Face Pose	Face Pose MLP	93%
Gesture — Eye Gaze	MPIIGaze MLP	95%
Gesture — Hand	HaGRID YOLOv8n	91%
Gesture — Fusion	Multimodal Fusion MLP	92%
Audio Analysis	wav2vec2-large-superb-er	76%
Sentiment Analysis	MobileNetV2 (FER2013)	73%



3.5.3 Discussion of Results

The results show that most modules achieve high performance, especially those based on transformer models and hybrid architectures:

1. The **CV-related modules** (classification, recommendation, and analysis) achieve the highest accuracy, exceeding 90%, confirming the effectiveness of NLP-based models in recruitment tasks.
2. The **gesture analysis module** performs well, particularly the MPIIGaze model, which reaches 95% accuracy for gaze estimation.
3. The **audio and sentiment analysis modules** show lower accuracy (76% and 73%), mainly due to the complexity and variability of emotional and acoustic data.

Note

Overall, these results demonstrate that combining multiple AI techniques leads to a robust and complete recruitment system capable of analyzing candidates from different perspectives.

3.6 Conclusion

This evaluation confirms the effectiveness of the **TalentSphere** architecture. The hybrid and multimodal approach allows the system to achieve strong performance across different tasks, including text, image, and audio analysis, making it suitable for real-world intelligent recruitment applications.

Deployment and Real-Time Testing

4.1 Introduction

This chapter focuses on the deployment and real-time testing of the **TalentSphere** platform. It describes how the different AI modules were integrated into a fully functional web-based system, allowing real-time interaction between users and the models.

The objective is to evaluate the system in a realistic environment, where candidates and recruiters can directly use the platform to test CV analysis, job matching, recommendation, and behavioral assessment modules.

4.2 Real-Time System Testing

To validate the performance of the system, the **TalentSphere** platform was deployed as a full-stack web application. This allows all AI models to be accessed through an interactive interface developed using React.js, Flask APIs, and NestJS backend services. Each module was tested using new and unseen data to simulate real recruitment scenarios.

4.2.1 User Dashboard Interfaces

The **TalentSphere** platform provides two main dashboards designed for different user roles: candidates and recruiters.

Candidate Dashboard — The candidate dashboard allows users to:

1. Upload and manage their CV



2. Access job recommendations
3. Receive CV analysis and feedback
4. Participate in AI-based interview simulations

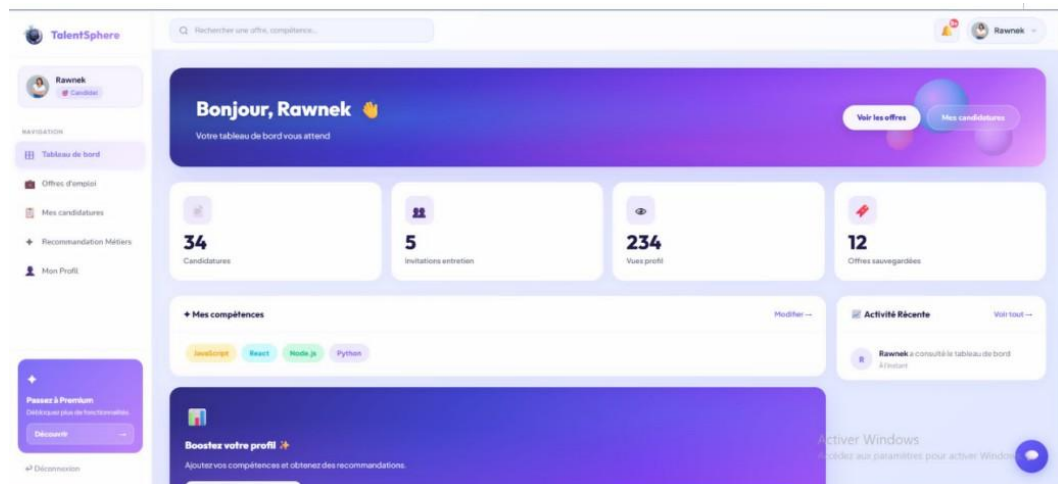


Figure 4.1. Candidate Dashboard Interface

Recruiter Dashboard — The recruiter dashboard allows users to:

1. Post and manage job offers
2. View and filter candidate applications
3. Access candidate evaluations and reports
4. Analyze interview results and final scores

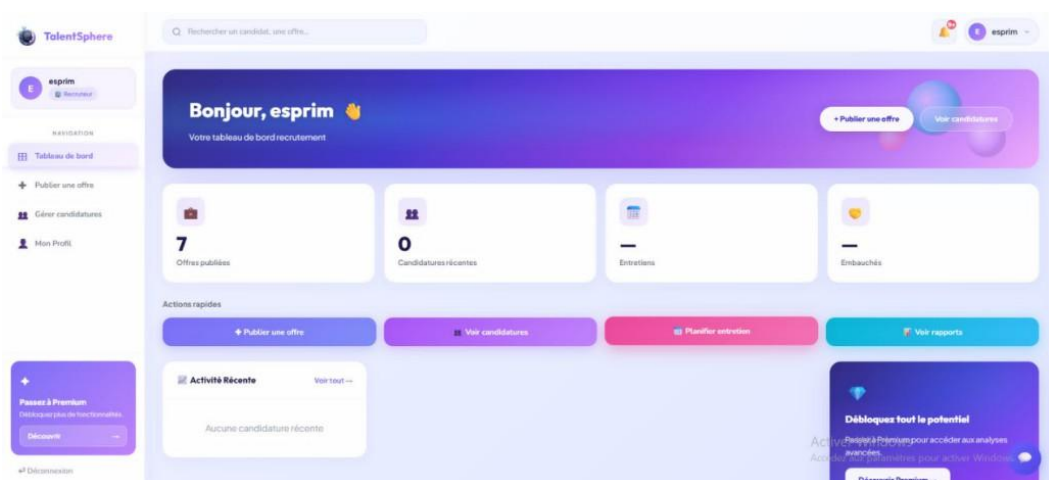


Figure 4.2. Recruiter Dashboard Interface



4.2.2 CV Classification & Matching Test

In this step, users can upload a CV and view the predicted job category along with matching job offers. The system automatically analyzes the CV and provides classification results in real time.

Candidat	CV	Classification	Scores	Statut	Actions	Date
R Rawnek rawnekakren12021@gmail.com	—	—	Matching CV 18.79% Quiz 25% Final 22%	• Entretien	✓ Accepter ✗ Refuser 📅 Planifier 🗑️ Générer Final	01 mai 2026
R Rawnek rawnekakren12021@gmail.com	—	—	Matching CV 18.79% Quiz 0% Final 22%	• En attente	✓ Accepter ✗ Refuser 📅 Planifier 🗑️ Générer Final	01 mai 2026
R Rawnek rawnekakren12021@gmail.com	—	—	Matching CV 18.79% Quiz 25% Final 22%	• Entretien	✓ Accepter ✗ Refuser 📅 Planifier 🗑️ Générer Final	01 mai 2026
R Rawnek rawnekakren12021@gmail.com	—	—	Matching CV 18.79% Quiz 25% Final 22%	• Entretien	✓ Accepter ✗ Refuser 📅 Planifier 🗑️ Générer Final	30 avr. 2026

Figure 4.3. CV Classification and Job Matching Interface — Real-Time Test

4.2.3 CV Analysis & Guidance Module Test

This module analyzes the content of the CV and generates personalized feedback to help candidates improve their profiles. The system highlights missing sections, weak skills, and improvement suggestions.

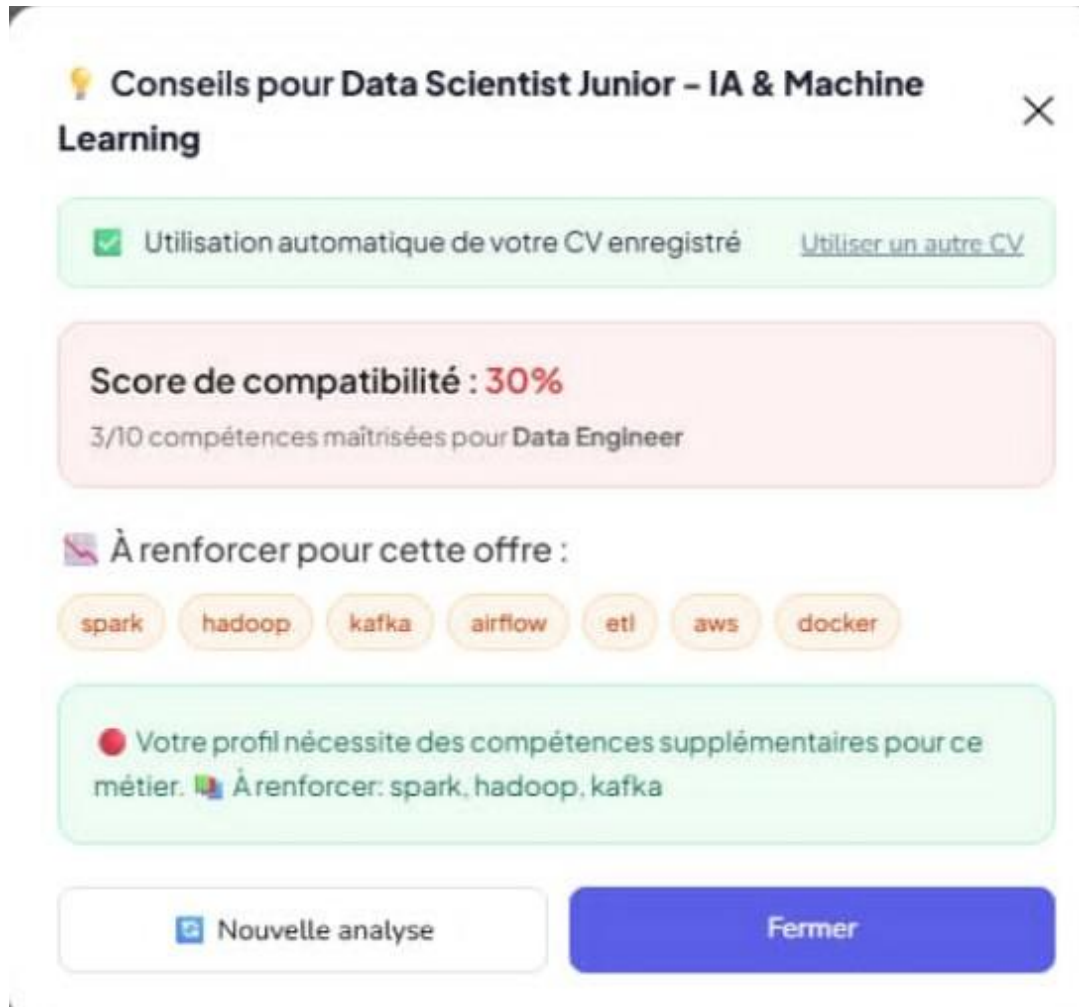


Figure 4.4. CV Analysis & Improvement Suggestions Output

4.2.4 Career Recommendation Module Test

In this test, the system recommends suitable job positions and career paths based on the candidate's CV and extracted skills. The recommendation engine combines TF-IDF, BERT embeddings, and skills matching.



Recommandation de Métiers

Uploade ton CV pour découvrir les **3 offres** qui matchent le mieux.

☒ Utilisation automatique de votre CV enregistré dans le profil [Utiliser un autre CV](#)

Fichier CV (PDF)

Aucun fichier choisi

RAOUNEK AKREMI
Tél. : +216 29 448 768 | E-mail : AKREMI.Raounek@esprim.tn | Monastir, Tunisie
LinkedIn : rawnek-akremi

PROFIL
Étudiante en 1er cycle ingénieur à l'École Supérieure Privée d'Ingénierie de Monastir, passionné par le développement logiciel, l'électronique embarquée et les systèmes inte...

Secteur: Tous les secteurs

☒ 3 offres trouvées

Développeur Full Stack - EdTech 54% match
Entreprise ** Mid

Data Scientist Junior - IA & Machine Learning 37% match
Entreprise ** Junior

Figure 4.5. Career Recommendation Results Interface

4.3 Conclusion

This chapter demonstrated the successful deployment of the **TalentSphere** platform and its real-time testing in a web environment. The system allows users to interact with multiple AI-powered modules, including CV analysis, job matching, recommendation, and multimodal interview evaluation.

The results confirm that the platform is capable of processing real-world data efficiently and providing meaningful insights for both candidates and recruiters. This makes **TalentSphere** a complete and intelligent recruitment solution that can be further improved and extended in future versions.

General Conclusion and Perspectives

This project led to the development of **TalentSphere**, an intelligent recruitment platform designed to automate and improve the hiring process using artificial intelligence. Unlike traditional recruitment systems, **TalentSphere** combines multiple AI techniques to analyze candidates from different perspectives, including CV content, skills, behavior, speech, and facial expressions.

Throughout this work, we covered the full pipeline: data preparation, model design, training, evaluation, and real-time deployment. The system integrates several specialized models such as XGBoost, RoBERTa, BERT-based recommendation systems, and multimodal deep learning models for gesture, audio, and sentiment analysis. These models work together to generate a complete and structured evaluation of candidates.

The results show that the platform is effective in providing accurate CV classification, relevant job recommendations, detailed CV feedback, and reliable behavioral analysis. The final output, generated as a recruiter report in PDF format, offers a global and objective overview of each candidate, helping recruiters make faster and more informed decisions.

Perspectives

Despite the good performance achieved, **TalentSphere** can be further improved in several directions:

1. **Cloud deployment:** Deploying the platform on cloud infrastructure for large-scale real-world usage and improved scalability.
2. **Multilingual support:** Adding support to handle CVs and interviews in multiple languages, making the platform globally accessible.
3. **Authentication extension:** Extending the system with facial recognition-based authentication for secure user login and access control.